

VIKASH KUMAR SINGH

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PROFESSIONAL SUMMARY

Software Developer with experience building scalable backend systems and modern web applications using Node.js, TypeScript, React.js, and MongoDB. Skilled in REST API development, database optimization, cloud deployment, and AI-powered applications using LangChain and LangGraph. Experienced in developing high-performance systems for EdTech and construction automation platforms.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Backend: Node.js, Express.js, Socket.io, JWT

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Databases: MongoDB, MySQL, PostgreSQL

Cloud & DevOps: Docker, AWS EC2, EBS, Route 53, Ubuntu, Github Action(CI/CD)

AI Technologies: LangChain, LangGraph, RAG, AI Agents

EXPERIENCE

Software Developer | Civil Guruji Pvt Ltd

August 2023 – Present

- Engineered scalable REST APIs using Node.js and TypeScript for authentication, course management, attendance systems, project management, and user engagement features.
- Built responsive frontend applications using React.js and Next.js for the CivilGuruji EdTech ecosystem and construction automation platforms.
- Optimized MongoDB and SQL database schemas, queries, indexing strategies, and aggregation pipelines for high-volume applications, improving backend scalability and reducing API response times by 30%.
- Developed Offline Resource Management platform that automated attendance tracking, assignment submissions, and centralized learning material distribution for offline learners.
- Developed web based BOQ, BOM, and Bar Bending Schedule (BBS) calculation systems to streamline construction estimation and documentation workflows.
- Managed deployment workflows and production infrastructure using Docker, Ubuntu, AWS EC2, EBS, Route 53 and Github Action.

PROJECTS

MYOPENCLAW

[github link](#)

- Built an AI-driven file management agent using TypeScript, LangGraph, LangChain, and Ollama, translating natural-language requests into executable file system operations.
- Architected a stateful workflow engine with planner, validator, and executor nodes, leveraging Redis checkpointing for interruption handling and workflow recovery.
- Developed multi-channel user interfaces through Telegram Bot (Telegraf) and CLI integrations, enabling automated file and folder management with structured validation and secure workspace isolation.

EDUCATION

B.Tech | Guru Ghasidas Vishwavidyalaya, Bilaspur

CGPA: 8.15 | 2019 – 2023

XII (CBSE) | DAV Public School, Dhori

82% | 2018